

Statistical Mechanics

20 marks

Quiz

45 minutes

- (1) Using the method of Lagrange multipliers, derive the canonical distribution by maximizing entropy under appropriate constraints. [5]
- (2) Derive an expression for energy fluctuations in the canonical ensemble and show how it depends on heat capacity. [5]
- (3) Starting from the definition of the partition function for an ideal gas of N identical particles confined in a volume V at temperature T , derive expressions for the internal energy, Helmholtz free energy, entropy, and heat capacity. [4]
- (4) Consider a system of N interacting identical particles with potential energy $U(\{r\})$. Write the expression for the probability of a microstate of this system in (a) the isothermal-isobaric ensemble and (b) the μVT ensemble. [2]
- (5) Determine the number of accessible microstates for an ideal gas of N identical particles in the microcanonical ensemble. [4]